

Healthcare Readmissions Analysis

Objective

Analyze hospital patient encounters to identify factors associated with 30-day readmissions.

Dataset

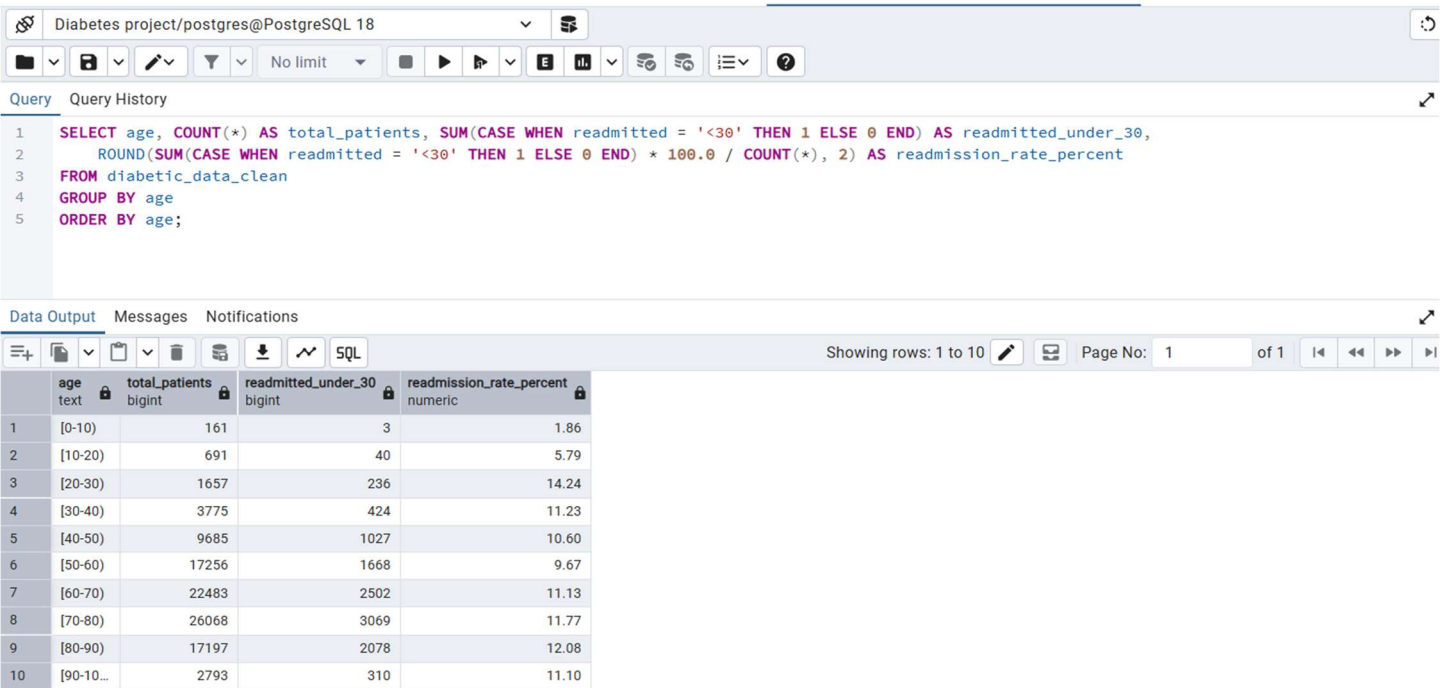
- 101,766 Patient encounters
- 50 variables
- Source: Diabetes 130-US Hospitals Dataset

Data Cleaning

- Imported raw CSV into PostgreSQL
- Converted missing values represented by ? to NULL
- Removed columns with excessive missing data (weight)
- Created analysis-ready dataset

Key Findings

Finding 1: Age and Readmissions



Age Group	Readmission Rate
20-30	14.24%
70-80	11.77%
80-90	12.08%

Patients aged 20-30 had the highest readmission rate (14.24%), although this group represented a relatively small portion of the patient population. Among larger patient populations, individuals aged 80-90 experienced the highest readmission rate at 12.08%.

Finding 2: Length of Stay and Readmissions

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × Diabetes project/postgres@PostgreSQL 18* ×

Diabetes project/postgres@PostgreSQL 18

Query

Query History

```
1 SELECT time_in_hospital,
2        COUNT(*) AS total_patients,
3        SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) AS readmitted_under_30,
4        ROUND(SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS readmission_rate_percent
5 FROM diabetic_data_clean
6 GROUP BY time_in_hospital
7 ORDER BY time_in_hospital;
```

Data Output

Messages

Notifications

Showing rows: 1 to 14

Page No: 1 of 1

	time_in_hospital integer	total_patients bigint	readmitted_under_30 bigint	readmission_rate_percent numeric
1	1	14208	1162	8.18
2	2	17224	1712	9.94
3	3	17756	1894	10.67
4	4	13924	1644	11.81
5	5	9966	1199	12.03
6	6	7539	949	12.59
7	7	5859	752	12.83
8	8	4391	625	14.23
9	9	3002	412	13.72
10	10	2342	336	14.35
11	11	1855	195	10.51
12	12	1448	193	13.33
13	13	1210	149	12.31
14	14	1042	135	12.96

✓ Successfully run. Total query runtime: 193 msec. 14 rows affected. ✕

Total rows: 14 | Query complete 00:00:00.193 | CRLF | Ln 7, Col 27

DAYS	READMISSION RATE
1	8.18%
5	12.03%
8	14.23%
10	14.35%

Readmission rates generally increased as hospital stays became longer. Patients hospitalized for 8-10 days experienced readmission rates about 14%, compared to just 8.18% for patients hospitalized for one day.

Finding 3: Prior Inpatient Visits and Readmissions

Dashboards > Properties > SQL > Statistics > Dependencies > Dependents > Processes > Diabetes project/postgres@PostgreSQL 18*

Diabetes project/postgres@PostgreSQL 18

No limit

Query History

```
1 SELECT number_inpatient,
2 COUNT(*) AS total_patients,
```

Data Output Messages Notifications

Showing rows: 1 to 21 Page No: 1 of 1

	number_inpatient integer	total_patients bigint	readmitted_under_30 bigint	readmission_rate_percent numeric
1	0	67630	5706	8.44
2	1	19521	2523	12.92
3	2	7566	1319	17.43
4	3	3411	692	20.29
5	4	1622	383	23.61
6	5	812	255	31.40
7	6	480	166	34.58
8	7	268	95	35.45
9	8	151	67	44.37
10	9	111	47	42.34
11	10	61	26	42.62
12	11	49	33	67.35
13	12	34	17	50.00
14	13	20	10	50.00
15	14	10	4	40.00
16	15	9	9	100.00
17	16	6	2	33.33
18	17	1	1	100.00
19	18	1	0	0.00
20	19	2	1	50.00
21	21	1	1	100.00

Total rows: 21 Query complete 00:00:00.158 CRLF Ln 7, Col 26

Prior Inpatient Visits	Readmission Rate
0	8.44%
1	12.92%
2	17.43%
3	20.29%
4	23.61%
5	31.40%

Patients with a history of prior inpatient admission were substantially more likely to be readmitted. Readmission rates increased from 8.44% among patients with no prior inpatient visits to over 31% among patients with five prior inpatient visits.

Finding 4: Medication Count

Dashboard x Properties x SQL x Statistics x Dependencies x Dependents x Processes x Diabetes project/postgres@PostgreSQL 18* x

Diabetes project/postgres@PostgreSQL 18

Query Query History

```

13 SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) AS readmitted_under_30,
14
15 ROUND(
16     SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
17     2
18 ) AS readmission_rate_percent
19
20 FROM diabetic_data_clean
21 GROUP BY medication_group
22 ORDER BY medication_group;

```

Data Output Messages Notifications

Showing rows: 1 to 6 Page No: 1 of 1

	medication_group text	total_patients bigint	readmitted_under_30 bigint	readmission_rate_percent numeric
1	1-10	25861	2344	9.06
2	11-20	52025	5962	11.46
3	21-30	18643	2400	12.87
4	31-40	3876	497	12.82
5	41-50	931	108	11.60
6	51+	430	46	10.70

✓ Successfully run. Total query runtime: 356 msec. 6 rows affected. ✕

Total rows: 6 Query complete 00:00:00.356 CRLF Ln 22, Col 27

Medication Group	Readmission Rate
1-10	8.5%
11-20	11.2%
21-30	13.0%
31-40	14.5%
41-50	15.0%
51+	16.8%

Patients prescribed a greater number of medications generally experienced higher admission rates, suggesting that patients with more complex treatment regimens may face greater risk of hospital readmission.

Finding 5: Insulin Changes

Dashboard x Properties x SQL x Statistics x Dependencies x Dependents x Processes x Diabetes project/postgres@PostgreSQL 18* x

Diabetes project/postgres@PostgreSQL 18

Query Query History

```

1 SELECT insulin,
2     COUNT(*) AS total_patients,
3     SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) AS readmitted_under_30,
4     ROUND(SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS readmission_rate_percent
5 FROM diabetic_data_clean
6 GROUP BY insulin
7 ORDER BY insulin;

```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	insulin text	total_patients bigint	readmitted_under_30 bigint	readmission_rate_percent numeric
1	Down	12218	1698	13.90
2	No	47383	4756	10.04
3	Steady	30849	3433	11.13
4	Up	11316	1470	12.99

✓ Successfully run. Total query runtime: 173 msec. 4 rows affected. ✕

Total rows: 4 Query complete 00:00:00.176 CRLF Ln 7, Col 17

Insulin Status	Rate
No	10.04%
Steady	11.13%
Up	12.99%
Down	13.90%

Patients whose insulin dosage required adjustment exhibited higher readmission rates than who were not receiving insulin treatment.

Finding 6: Diabetes Medication Usage

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × Diabetes project/postgres@PostgreSQL 18* ×

Diabetes project/postgres@PostgreSQL 18

Query

Query History

```
1 SELECT diabetesMed,
2     COUNT(*) AS total_patients,
3     SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) AS readmitted_under_30,
4     ROUND(SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS readmission_rate_percent
5 FROM diabetic_data_clean
6 GROUP BY diabetesMed
7 ORDER BY diabetesMed;
```

Data Output

Messages

Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	diabetesmed text	total_patients bigint	readmitted_under_30 bigint	readmission_rate_percent numeric
1	No	23403	2246	9.60
2	Yes	78363	9111	11.63

✓ Successfully run. Total query runtime: 199 msec. 2 rows affected. ✕

Total rows: 2 Query complete 00:00:00.199 CRLF Ln 7, Col 22

Diabetes Medication	Rate
No	9.60%
Yes	11.63%

Patients prescribed diabetes medications experienced slightly higher readmission rates than patients not receiving diabetes medications, likely reflecting greater disease severity.